Implementation and Comparison of Six Pedestrian Trajectory Prediction Models

COMP5331 Knowledge Discovery in Databases | Fall 2026

Project type: Implementation

Suggested topic: Topic 6 — Mining Spatial and Temporal Data

Group number: 7

Group information and declarations

Each member should complete their own section below.

Member 1

Student name: LU Yichong

Student ID: 21350771

Research or FYP supervisor: Qifeng Chen

Own research or FYP topic: World Models

Difference from own research or FYP: The goal of "World Models" is to represent the world, which is totally different with "Pedestrian Trajectory Prediction".

Declaration for each member: I declare that this project will be carried out solely for COMP5331. The work submitted for assessment will not be submitted for another course or counted as a deliverable of my research or FYP project. All external papers, datasets, software, and other assistance will be acknowledged.

Member 2

Student name: Zongmin Zhang

Student ID: 21041605

Research or FYP supervisor: Prof. Chaojian Li (FYT supervisor) and Prof. Lixue Cheng (research supervisor and FYT co-supervisor)

Own research or FYP topic: own research topic: AI for Science; FYT topic: Embodied Multi-Agent System for High-Stakes Environment Safety: Transferring Autonomous Driving Architectures to Edge-Deployed Labs

Difference from own research or FYP: My research focuses on AI for Science, and my FYP develops an embodied multi-agent safety system for edge-deployed laboratories, combining physical sensing, evidence-grounded chemical-procedure review, mobile-edge deployment, runtime assurance, and hardware-in-the-loop evaluation. This COMP5331 project instead implements and compares pedestrian trajectory-prediction models on the ETH/UCY benchmark. Although both projects involve multi-agent and spatiotemporal reasoning, they differ in problem domain, data, objectives, implementation, experiments, and assessed deliverables.

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Member 3

Student name: Yifan WANG

Student ID: 20945256

Research or FYP supervisor: Prof. Qifeng Chen

Own research or FYP topic: Audio-Video Generation

Difference from own research or FYP: The goal of " Audio-Video Generation" is for entertainment, which is totally different with "Pedestrian Trajectory Prediction".

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Member 4

Student name: Cheung, Alan Pak To

Student ID: 21055723

Research or FYP supervisor: Arpit Narechania

Own research or FYP topic: Analyzing Text in Videos

Difference from own research or FYP: The goal of this project is to exctract text from the moive opening/ending credit and analysis the relationship and networks among crews of the moives, Which is different from Trajectory Prediction.

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Member 5

Student name: YAN Zhengyang

Student ID: 21320594

Research or FYP supervisor: Song Guo

Own research or FYP topic: Reinforcement Learning

Difference from own research or FYP: My own research is focus on reinfrocement learning implementation around robotics. My reserach has no any direct relations to KDD and time series.

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Member 6

Student name: LI Junhao

Student ID: 21352303

Research or FYP supervisor: Song Guo

Own research or FYP topic: Agentic Robotics

Difference from own research or FYP: My own research is focusing on agentic harness implementation around robotic problems. My reserach has no any direct relations to KDD and time series.

Declaration for each member: I declare that this project will be carried out solely for COMP5331. The work submitted for assessment will not be submitted for another course or counted as a deliverable of my research or FYP project. All external papers, datasets, software, and other assistance will be acknowledged. [Each member must verify this declaration before submission.]

Project description

Motivation and objective

We propose an implementation project comparing six pedestrian trajectory prediction models on a common benchmark. Predicting future movement requires mining temporal patterns in individual trajectories and spatial interactions among nearby pedestrians. Turning, crowd interactions, and imperfect observations make this task challenging. Our objective is to examine how different modeling choices affect prediction accuracy, robustness, motion consistency, and computational cost.

DSPIGCN [1], a paper listed under Topic 6, will anchor the project. Its physics-informed graph architecture will be compared with an individual LSTM baseline, Social-LSTM [7], Social-GAN [8], Social-STGCNN [2], and Trajectron++ [3]. These implementations cover individual sequence modeling, social interaction modeling, probabilistic generation, graph-based prediction, and physical constraints. This breadth supports a comparative implementation study without requiring a new prediction algorithm.

Each of the six members will independently complete one model's entire experimental workflow. The project will combine six self-contained implementations and standardized results. The assessed contribution will include model implementation, reproducible evaluation, robustness experiments, and a critical comparison of the resulting evidence. We will distinguish our implementation work from reused reference code and will not assume that the newest model performs best.

Problem definition and model scope

Each model will receive pedestrian identifiers, timestamps, and two-dimensional observed positions in metric coordinates. Eight observed steps will be used to predict the next twelve steps. All models will use identical observation and prediction times and the same eligible evaluation trajectories. Inputs will be limited to trajectory information; map images and other additional modalities will be excluded. Model-specific coordinate transformations must be reversible, with predictions evaluated in the original metric coordinates.

The individual LSTM will be a deterministic encoder-decoder baseline that predicts future positions from one pedestrian's history without information from neighboring pedestrians. Its architecture and training loss will be stated explicitly as our baseline design. A simple constant-velocity predictor will be implemented alongside it as an inexpensive sanity check, rather than assigned as a separate member's main model.

Social-LSTM [7] will model neighboring pedestrians through social pooling of recurrent representations. Social-GAN [8] will use recurrent trajectory generation, a pooling mechanism, adversarial training, and a variety loss to produce multiple plausible futures. Social-STGCNN [2] will represent interactions through a spatiotemporal graph and use temporal prediction layers. Each model owner will reproduce the relevant core components and document departures from the paper or reference implementation.

Trajectron++ [3] will provide a graph-structured probabilistic comparator. We will use its pedestrian configuration with trajectory-derived states and no map input, documenting its dynamics and sampling settings. The implementation will be limited to pedestrian forecasting on the selected benchmark, rather than reproducing every modality and application supported by the original system.

DSPIGCN [1] will be implemented with displacement and velocity streams, temporal attention, graph convolution, physics coupling, and anchor-based sampling. Its owner will verify the associated reconstruction, likelihood, and physical-consistency losses against the paper and available artifact. Small examples will check dimensions, coordinate units, and derivative calculations before full training. Removing the velocity stream or physical regularizer may be examined as an optional extension after the required comparison is complete.

Data and experimental protocol

The benchmark will comprise ETH, Hotel, Univ, Zara1, and Zara2 from ETH/UCY [4, 5]. The DSPIGCN paper uses the label GC for the ETH and Hotel scenes; our documentation will identify the actual scene files. Before independent implementation begins, the group will fix the public data release, sampling interval, eligible trajectory identifiers, and split definitions in a short protocol. Each member will then independently prepare the data required by their model from that release.

Evaluation will use leave-one-scene-out testing. One scene will be held out and the others will provide training and validation data. Validation segments will be disjoint from training segments before observation windows are generated, with a boundary gap sufficient to prevent overlapping trajectories across the split. Learned normalization will use training data only. All six implementations must reproduce the agreed sample counts and identifiers; model-specific filtering must not silently change the evaluated population.

Each member will first complete a pilot fold, then train and evaluate their model across all five folds. The main comparison will target three training seeds per learned model and report per-scene results and variation across seeds. Hyperparameters and checkpoint selection will use validation data under documented training limits. Owners will record dependencies, configurations, data manifests, seeds, and checkpoints. Published results will be compared directly only when their preprocessing and evaluation conventions match our protocol.

Evaluation of accuracy and reliability

All owners will independently calculate average displacement error and final displacement error in meters. Probabilistic methods will generate twenty future samples per pedestrian and report both minimum and mean errors across samples, following an agreed aggregation rule. The deterministic LSTM and constant-velocity check will return one trajectory. Their results will be labeled separately so that a single prediction is not presented as having the same sampling budget as a best-of-twenty result.

For probabilistic models, we will also examine sample diversity and a sample-based energy score [6] over the future coordinate sequence. Minimum error indicates whether one sample covers the realized future; it does not establish the quality of the full distribution. Mean error, diversity, and energy score provide complementary evidence. We will avoid interpreting any one metric as proof of calibrated uncertainty.

Motion diagnostics will summarize speed, acceleration, abrupt changes, and close approaches between simultaneously predicted pedestrians. Thresholds will be fixed using training data and stated in the protocol. Close approaches will be treated as a spatial-consistency diagnostic, not a guarantee about collision avoidance. Each owner will also measure preprocessing, prediction, and sampling time under the same pedestrian count and hardware setting. Runtime measurement boundaries and warm-up procedures will be fixed in advance.

To prevent independently written evaluators from producing incompatible scores, the protocol will include a tiny example with known ADE/FDE values and explicit sampling and aggregation rules. Each owner will check their evaluator against this example before running the full experiments. Final tables will retain the model, scene, seed, sample count, and test-set identifier so that results can be audited without integrating model code.

Robustness tests and qualitative analysis

Every member will run the same perturbation conditions on their own model. Gaussian position noise will have standard deviations of 0, 0.05, 0.10, and 0.20 meters. Separate missing-observation conditions will hide the last one or two observed steps while leaving future targets unchanged. The protocol will specify the random generator, sample ordering, and seeds so that independently generated corruptions match. Missing suffixes will be filled using the same causal extrapolation rule based on the remaining observations.

Each owner will compare clean and perturbed prediction errors and create trajectory plots showing observed histories, ground-truth futures, and predictions. A fixed set of example identifiers will support visual comparison across models. Owners will also select failures from their own results and explain whether errors arise around turns, dense interactions, or corrupted observations. These analyses will be completed within each model's workflow; no member will wait for another person to supply an evaluation or visualization module.

Division of work among six members

Each member will own data preparation, model implementation, training, validation, clean and perturbed evaluation, runtime measurement, visualization, and the report section for one model. Members will work in separate directories or repositories with their own environments. The allocation below identifies the model and its particular analysis focus; it does not split shared pipeline stages across people.

Member 1 (Name: LU Yichong) — Individual LSTM. Complete the deterministic sequence-model baseline and the supplementary constant-velocity sanity check. Analyze when learning individual motion improves on simple extrapolation and where the absence of neighbor information limits prediction.

Member 2 (Name: WANG Yifan) — Social-LSTM. Complete social pooling, recurrent prediction, training, and all required experiments. Analyze behavior in crowded and turning trajectories and report implementation or reproduction discrepancies.

Member 3 (Name: Cheung, Alan Pak To) — Social-GAN. Complete generator, discriminator, pooling, sampling, and training. Analyze the relationship between best-sample accuracy, typical-sample error, and diversity, alongside the common robustness and runtime results.

Member 4 (Name: Zongmin Zhang) — Social-STGCNN. Complete graph construction, temporal prediction, training, and evaluation. Analyze accuracy and runtime across the agreed scenes and pedestrian counts, together with sensitivity to perturbed histories.

Member 5 (Name: LI Junhao) — Trajectron++. Complete the pedestrian configuration, data adaptation, training, probabilistic prediction, and evaluation. Document the chosen dynamics and sampling settings and analyze the resulting motion consistency and prediction distribution.

Member 6 (Name: YAN Zhengyang) — DSPIGCN. Complete the full physics-informed model, its training, and all required experiments. Analyze clean and corrupted-input performance and motion consistency. Optional local ablations will be undertaken only after the mandatory results are complete.

Coordination will be limited to agreeing on the experimental protocol at the start and combining results at the end, with questions raised when deviations or blockers occur. Each member will submit runnable code, configurations, checkpoints, exported predictions, a standard results table, figures, and their completed report section. A designated editor will assemble the sections and comparison table without rerunning other members' workflows. Shared GPU access will use a booking sheet. Actual contributions will be recorded individually.

Work plan and feasibility

Weeks one and two will cover protocol agreement, paper reading, environment setup, and each owner's data preparation. During weeks three and four, all six members will independently implement their models and finish a pilot fold. Weeks five and six will cover the main five-fold experiments. Weeks seven and eight will cover robustness tests, analysis, and report assembly. These milestones will be aligned with official presentation and submission dates.

Six independent implementations reduce software dependencies between members but increase total training and duplicated setup work. Initial planning assumes one shared GPU; each owner will measure pilot runtime before reserving training slots. Broad parameter searches and optional ablations will be omitted if necessary to prioritize the required comparison. Any reduction to the common folds or seed budget must be documented consistently and discussed with the instructor if it materially changes the scope. Authors' code may guide implementation and validation, with reused components explicitly acknowledged.

Expected deliverables

We will deliver six self-contained model packages, one agreed evaluation protocol, a combined comparison table, robustness results, trajectory figures, and a report. Every reported number will be traceable to an owner's saved configuration and output. The final discussion will assess tradeoffs in accuracy, uncertainty, robustness, physical consistency, and runtime, while distinguishing observed results from architectural explanations that the experiments cannot establish causally.

Papers to be read

[1] Runkang Guo, Bin Chen, Zhengqiu Zhu, Chen Gao, Yong Zhao, and Quanjun Yin. 2026. DSPIGCN: Dual-stream Physics-informed Graph Convolutional Network for Reliable Pedestrian Trajectory Prediction. KDD 2026, 383–394. Main implementation target from Topic 6. https://doi.org/10.1145/3770854.3780329

Course PDF: https://canvas.ust.hk/courses/72132/files/12591888

Author-linked software artifact: https://doi.org/10.5281/zenodo.18144387

[2] Abduallah Mohamed, Kun Qian, Mohamed Elhoseiny, and Christian Claudel. 2020. Social-STGCNN: A Social Spatio-Temporal Graph Convolutional Neural Network for Human Trajectory Prediction. CVPR 2020, 14424–14432. Implemented graph-based baseline. https://arxiv.org/abs/2002.11927

Reference code: https://github.com/abduallahmohamed/Social-STGCNN

[3] Tim Salzmann, Boris Ivanovic, Punarjay Chakravarty, and Marco Pavone. 2020. Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data. ECCV 2020, 683–700. Implemented probabilistic comparator using the pedestrian configuration. https://arxiv.org/abs/2001.03093

[4] Stefano Pellegrini, Andreas Ess, Konrad Schindler, and Luc Van Gool. 2009. You'll Never Walk Alone: Modeling Social Behavior for Multi-target Tracking. ICCV 2009, 261–268. Dataset and social-motion background.

[5] Alon Lerner, Yiorgos Chrysanthou, and Dani Lischinski. 2007. Crowds by Example. Computer Graphics Forum, 26(3), 655–664. UCY dataset background.

[6] Tilmann Gneiting and Adrian E. Raftery. 2007. Strictly Proper Scoring Rules, Prediction, and Estimation. Journal of the American Statistical Association, 102(477), 359–378. Background for probabilistic evaluation and the energy score. https://doi.org/10.1198/016214506000001437

[7] Alexandre Alahi et al. 2016. Social LSTM: Human Trajectory Prediction in Crowded Spaces. CVPR 2016. Implemented social recurrent baseline. https://doi.org/10.1109/CVPR.2016.110

[8] Agrim Gupta, Justin Johnson, Li Fei-Fei, Silvio Savarese, and Alexandre Alahi. 2018. Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks. CVPR 2018. Implemented generative comparator. https://arxiv.org/abs/1803.10892